

AR-Supported Online Piano Lessons with Bidirectional Visualization of Teacher and Student Hand Movements

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Figure 1: (a) AR piano lesson interface with overlaid remote hand movements. (b) Teacher-side setup visualizing the student's hands in real time. (c) Student-side setup visualizing the teacher's hands in real time.

Abstract

Online piano lessons have rapidly gained popularity in recent years. However, conventional online lessons often fail to convey detailed fingerings and hand movements effectively due to limited camera views. In this study, we propose an augmented reality (AR)-supported online piano lesson system that enables teachers and students in different locations to mutually visualize their hand performances in real time. This system allows intuitive understanding of finger movements and key operations, facilitating bidirectional feedback. A pilot study with a small number of participants revealed that mutual visualization can enhance comprehension and ease of instruction, while also posing challenges related to increased cognitive load. This work demonstrates the potential of AR technology for advancing online music education.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction design**; **Visualization**.

Keywords

Augmented Reality, Online Piano Lessons, Music Education, Bidirectional Visualization, Human-Computer Interaction

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1 Introduction

Since the COVID-19 pandemic, online lessons have expanded across both formal and extracurricular education. In music education, online piano lessons have grown in popularity, offering reduced commuting and flexible instructor choice. However, studies have reported challenges such as limited interactivity and poor visibility of hand movements [Nagasawa and Inoue 2020]. In typical video lessons, students place a camera beside the keyboard and must look away from the keys, making it difficult to follow finger movements and hindering feedback.

To address these issues, we propose a bidirectional AR-supported piano lesson system that enables teachers and students to visualize and share hand performances in real time. This allows teachers to monitor playing and students to compare their movements directly with the teacher's demonstrations. We present the system overview and pilot study results, highlighting key benefits and challenges.

2 Related Works

Several studies have explored technology-assisted music learning. Matsui et al. [Matsui et al. 2018] proposed a remote lesson system with collaborative annotation and camera switching. Applications such as PianoVision [PianoVision Team 2024] leveraged AR/VR to overlay notes on keyboards for intuitive, score-free practice. Other research visualized teachers' hand movements on students'

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keyboards [Yamashita and Masuko 2024], but these were mostly one-directional. Labrou et al. [Labrou et al. 2023] developed an MR application showing 3D traces of teachers' hands aligned with scores, helping novices replicate gestures and demonstrating the potential of immersive visualization.

Building on these studies, our work advances a bidirectional AR system enabling teachers and students to observe each other's hands in real time.

3 Methodology

3.1 System Overview

The proposed system enables bidirectional AR interaction between a teacher and a student during piano lessons. Each participant's hand is captured with an RGB camera and transmitted over a local network. The video stream is processed in Unity, which performs background removal to extract hands. These are then superimposed in real time on the participant's keyboard view through a head-mounted display (HMD; Meta Quest 3) using its passthrough API, allowing both users to perceive the overlaid hands within their physical environment. Figure 1(a) shows an example interface, with the student's hands overlaid on the teacher's from the teacher's perspective.

For alignment, the system currently uses manual calibration: before the lesson, users adjust scale and position in Unity to match displayed hands with their keyboard, ensuring finger movements correspond to correct keys.

To maintain synchronization between visual and auditory feedback, Open Sound Control (OSC) exchanges note-on/off messages between teacher and student devices, allowing both to hear synchronized audio with the visual overlay.

The overall configuration (Figure 2) includes (1) hand capture, (2) network transmission, (3) Unity-based AR overlay, and (4) OSC-based audio synchronization, enabling real-time bidirectional feedback.

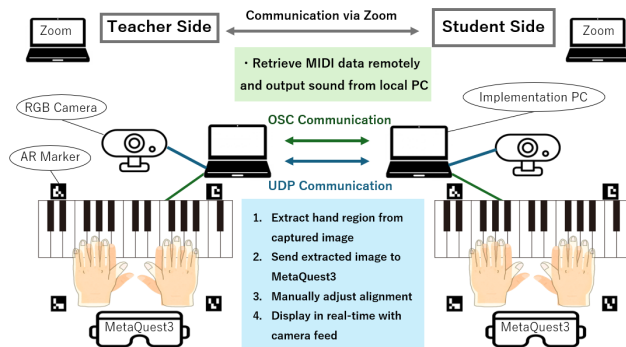


Figure 2: System configuration of the AR-based piano lesson showing real-time hand capture, transmission, and overlay.

3.2 Pilot Study Setup

To examine feasibility, we conducted a pilot study with six graduate students—three teachers and three students—forming three online lesson pairs. Teachers had over three years of piano training, while

students had little or none. The teacher setup during lessons is shown in Figure 1(b), and the student setup in Figure 1(c). Each pair experienced two lesson types in sequence: (1) a conventional one-way system and (2) the proposed bidirectional AR system, each lasting about 15 minutes. The instructional task used beginner pieces, “Twinkle, Twinkle, Little Star” and “Silent Night,” with about 10 minutes on the first four measures and 5 minutes on the next four when time allowed. All lessons were conducted online.

3.3 Observations And Results

The pilot compared conventional one-way lessons with bidirectional AR lessons. Lesson 2 allowed teachers and students to visualize each other's hands in real time. Five of six participants found it more effective for beginners, suggesting mutual visualization improves instructional clarity and engagement.

Students reported that following the teacher's finger movements aided understanding of key positions and finger use. Some noted overlapping fingers and slight misalignment made observation difficult, suggesting that selectively displaying relevant hands or fingers could improve clarity. Teachers valued observing students' hands, though misalignment and occasional obstruction posed challenges. Overall, teachers found the AR system engaging, with better hand alignment needed for practical use.

These results suggest that while the system enhances learning and engagement, refinement in selective display and alignment remains necessary.

4 Discussion and Future Work

The pilot study demonstrated that the proposed AR system can enhance clarity and immediacy of instruction, improving visibility of hand movements. Some participants noted that overlaying both hands created visual clutter and made focus difficult. Others suggested better hand alignment and easier calibration. These findings indicate the value of adaptive visualization, such as selectively displaying the most relevant hand or emphasizing key fingers to reduce cognitive load and aid learning.

Future work will refine visualization and alignment methods and include longer-term studies to evaluate usability and effectiveness. These efforts aim to advance AR-based piano education toward practical remote deployment. Beyond piano, the approach could extend to other hand-based skills, suggesting broader AR applications in education.

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